PMOS: An Integrated Biological Knowledge Base

Project Progress and Work Completed

Current Progress

1. Understanding of PMOS and Problem Identification

Poly Endocrine Metabolic Ovarian Syndrome (PMOS) is being approached in this project as a biological information integration problem involving interconnected endocrine, metabolic, and ovarian processes. Understanding the biological context was the first step before deciding what information should be collected and how the different biological entities could be related.

The project focuses on organizing relevant information such as genes, proteins, hormones, biomarkers, biological pathways, therapeutics, and supporting scientific evidence into a single searchable knowledge base.

For example, **CYP17A1** can be considered as a representative gene because of its role in steroid hormone related processes. Similarly, **testosterone** represents a hormone related to androgen activity, while **Anti Mullerian Hormone** can be considered as a reproductive biomarker. **Metformin** provides an example of a therapeutic compound that can be studied through its chemical and biological information.

The aim is to connect such information so that a user can explore related biological information without having to manually search several independent resources.

Outcome: The biological scope of the project was defined and the major information categories required for the knowledge base were identified.

2. Identification of Supporting Biological Databases

Existing biological databases were examined to determine which resources could provide reliable information for each component of the proposed knowledge base. The sources were selected based on their biological coverage and usefulness for cross referencing.

Database	Information to be Used
NCBI Gene	Gene identifiers, gene names, descriptions, and functional information
UniProt	Protein names, functions, annotations, and identifiers
Reactome	Biological pathways and pathway associated genes
Gene Ontology	Molecular functions and biological processes
PubChem	Chemical and therapeutic information
Open Targets	Gene and disease associations and supporting evidence
PubMed	Scientific publications and research evidence

These resources provide complementary information rather than serving as independent datasets. For example, NCBI Gene can provide information for **CYP17A1**, UniProt can provide the corresponding protein information, and Reactome can provide pathway information related to steroid hormone biosynthesis.

PubChem can be used for compounds such as **metformin**, while PubMed can provide supporting research publications. Open Targets can be used to examine gene and disease associations.

Outcome: The major external biological databases were identified and mapped to the types of information they can contribute to the proposed knowledge base.

3. Biological Entity Identification and Mapping

The next step was to identify how the different types of biological information could be related. This is important because the proposed system is intended to provide connected biological information rather than isolated records.

The preliminary relationship model is:

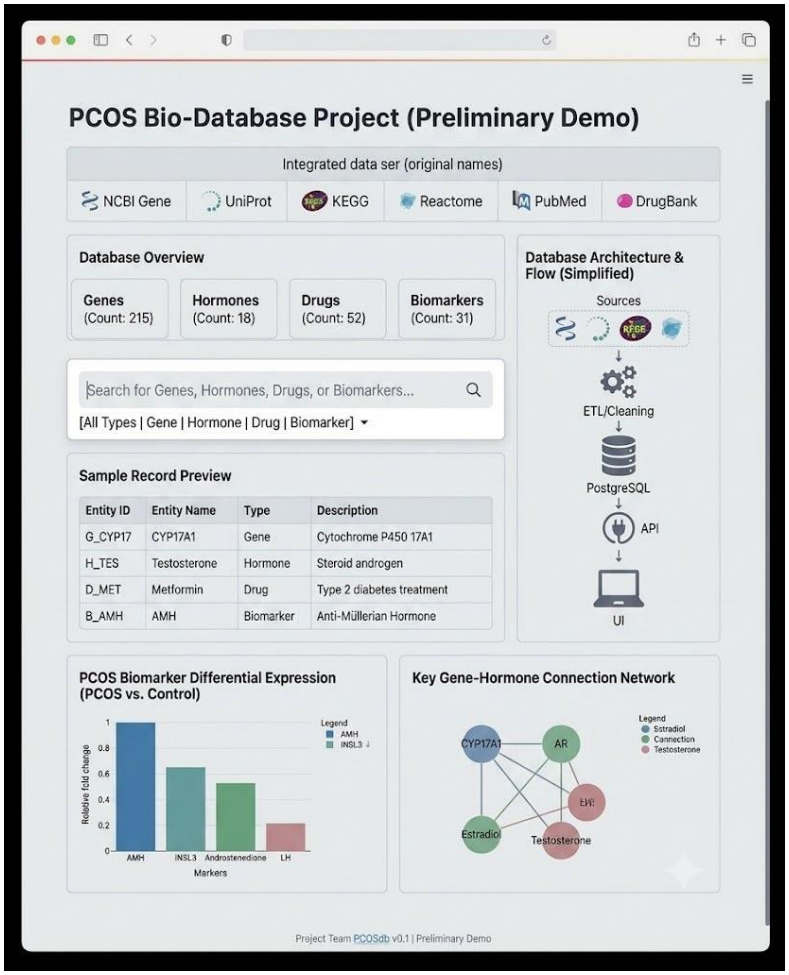
Gene → Protein → Pathway

Additional relationships considered include:

Hormone → Biological Process → Supporting Evidence

and

Therapeutic → Biological Target → Pathway



For example, **CYP17A1** can be connected to its corresponding protein and steroid hormone related pathways. **Testosterone** can be viewed in relation to androgen related biological processes, while **metformin** can be considered in relation to its therapeutic and metabolic context.

Outcome: The major biological entities and their preliminary relationships were mapped, providing a basis for organizing the integrated database.

4. Preliminary Data Curation

A preliminary set of biological records was identified and organized to test the proposed data structure. This stage was used to understand what information should be retained from each source and how it could be represented consistently.

Representative entities considered during the initial curation include:

- **Gene:** CYP17A1
- **Hormone:** Testosterone
- **Biomarker:** Anti Mullerian Hormone
- **Therapeutic:** Metformin
- **Pathway:** Steroid hormone biosynthesis

The initial records were examined for identifiers, names, descriptions, source information, and relevant biological relationships.

The purpose of this stage was to establish a representative sample before expanding the dataset.

Outcome: A preliminary collection of representative biological records was prepared and used to check whether the proposed data categories could be organized consistently.

5. Data Standardization and Identifier Mapping

Information obtained from different biological databases may use different names or identifiers for the same biological entity. Therefore, a standardization strategy was considered before combining the information.

For example, **CYP17A1** can be represented using its gene symbol, NCBI Gene identifier, and corresponding UniProt protein identifier. Maintaining these identifiers helps connect records from different sources without treating the same biological entity as separate records.

The preliminary curation process is:

Data Collection → Cleaning → Identifier Mapping → Validation

Source information will also be retained so that the origin of a biological record can be identified when required.

Outcome: A preliminary data standardization and identifier mapping strategy was established to support consistent integration of information from multiple biological resources.

6. Preliminary Database Structure

Based on the information identified so far, a relational structure was planned to organize the major biological entities while maintaining their relationships.

Entity	Example Information
Genes	Gene symbol, name, identifier, description
Proteins	Protein name, identifier, function
Hormones	Hormone name, biological role
Biomarkers	Biomarker name, evidence, description
Pathways	Pathway name, identifier, description
Therapeutics	Drug name, mechanism, identifier
Publications	PMID, title, year, source

The structure is intended to keep different biological entities organized while allowing relationships to be maintained between them.

For example, a single gene may be associated with multiple pathways, while a pathway may contain multiple genes. Similar relationships can be maintained between biological entities and supporting publications.

Outcome: A preliminary relational database model was established to organize the biological entities and their relationships.

7. Preliminary User Interface Prototype

A preliminary user interface was designed to demonstrate how users could interact with the proposed knowledge base.

The initial interface concept includes:

- Search by genes, hormones, drugs, and biomarkers
- Overview of available biological entities
- Display of representative biological records
- Identification of supporting data sources
- Preliminary biological relationship visualization

For example, a search for **CYP17A1** can be designed to display its gene information together with related protein and pathway information. Similarly, a search for **metformin** can display its chemical information and the relevant biological context stored in the database.

The current interface is being used as a prototype to validate how the information could be presented to users.

Outcome: A preliminary interface prototype was developed to demonstrate the proposed search workflow and organization of biological information.

8. Current Progress Summary

The work completed so far can be summarized as follows:

- Project scope and biological objectives defined
- Relevant biological databases identified
- Biological entities and preliminary relationships mapped
- Representative biological records identified
- Data fields for the major entities identified
- Initial data curation performed
- Data standardization strategy established
- Preliminary relational database structure designed
- Preliminary user interface prototype developed

Overall Outcome: Approximately 40% of the project foundation has been completed. The biological scope, supporting resources, representative data, data organization, preliminary database structure, and user interface have been established for the next stage of implementation.

9. Planned Next Phase

The next stage will focus on expanding the preliminary work and developing the functional components of the knowledge base.

The planned work includes:

- Expanding the curated biological dataset
- Collecting additional relevant genes, hormones, biomarkers, pathways, and therapeutics
- Connecting information from the selected biological resources
- Populating the database with validated records
- Implementing search and filtering
- Adding supporting publication information
- Establishing and validating biological relationships
- Improving the biological visualizations
- Testing the integrated system

Expected Outcome: A functional prototype of the PMOS biological knowledge base that allows users to search and explore interconnected biological information from multiple established resources.

References

- National Center for Biotechnology Information (NCBI). *CYP17A1: Cytochrome P450 Family 17 Subfamily A Member 1*. NCBI Gene, Gene ID 1586.
- UniProt. *CYP17A1: Steroid 17-alpha-hydroxylase/17,20-lyase*. UniProt accession P05093.
- Reactome. *Steroid hormone biosynthesis and related steroidogenic pathways*. Homo sapiens.
- Gene Ontology Consortium. *Gene Ontology: the unified framework for the functional annotation of genes and gene products*.
- PubChem. *Metformin*. PubChem Compound ID 4091.
- Open Targets Platform. *Gene and disease association evidence for human biological targets*.
- PubMed. *CYP17A1 rs743572 polymorphism and polycystic ovary syndrome: systematic review and meta-analysis*.